GM ASEP / A3 MANUAL DRIVE TRAIN & AXLES / UNIT 2

TRANSMISSION & TRANSAXLE INTERNALS

Gearset theory, synchronizer operation, and the shift-symptom logic covered between the Unit 1 clutch quiz and the Unit 2 quiz.

Prepared September 11, 2026 **Source** class notes, chapter 5 & instructor discussion

Scope gearset & synchronizer only — clutch (Unit 1) and differentials/axles (Unit 3) covered separately

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01 GEAR THEORY & POWER FLOW

Think of the transmission as a **selective torque multiplier**: low gears trade speed for torque, high gears trade torque for speed. Every ratio is that trade — you never get both for free.

1.1 GEAR RATIO BASICS

- **Reduction** — output turns slower than input, torque increases (numerically higher ratio, e.g. 3.5:1).
- **Direct drive** — 1:1, input speed equals output speed, no multiplication.
- **Overdrive** — output turns faster than input, torque drops below engine output (ratio under 1:1, e.g. 0.8:1).

Ratio math: driven teeth ÷ drive teeth. A **`10:7:20`** three-gear train still reduces to a **`2:1`** overall ratio — the idler's own tooth count never enters the final number.

> RULE An **idler gear** reverses rotational direction only. Even at a different diameter than the drive/driven gears, it does not change the ratio.

1.2 CONSTANT-MESH PRINCIPLE

Gear pairs stay **always meshed** — nothing slides tooth-to-tooth during a shift. What changes is which gear gets **locked to its shaft** by a synchronizer sleeve.

> MEMORIZE It is **never** a fixed gear meshed to a fixed gear — always fixed gear ↔ speed gear (a gear that freewheels on the shaft until locked). Only one gearset is locked at a time.

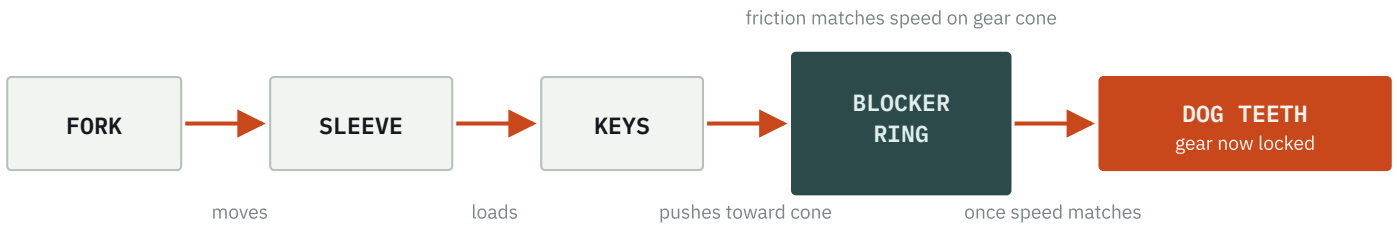
1.3 GEAR TYPE & LUBRICATION

Helical gears run quiet but generate **endthrust** (axial load) — shafts need thrust surfaces to control it. Most manual transmissions are **splash lubricated**: the countershaft dips into the sump, spins, and throws oil onto the mainshaft. There's no oil pump in a typical setup.

02 SYNCHRONIZER ASSEMBLY & SHIFT MECHANISM

This is the core Unit 2 topic. A complete synchronizer assembly is the **hub, sleeve, blocker/synchronizer ring(s), keys/struts, and springs**. The speed gear and shift fork work with it but are not part of the assembly itself.

PART	WHAT IT DOES
Hub	Splined solid to the shaft — never moves independently of it.
Sleeve	Slides axially on the hub's external splines; the shift fork moves it.
Speed gear	Freewheels on the shaft until locked; has a friction *cone* and side *dog/clutch teeth*.
Blocker ring	Friction ring with limited rotational play. Contacts the cone; while speeds differ, friction twists it just enough to misalign its teeth from the sleeve — physically blocking premature engagement.
Keys / struts	Spring-loaded, seated under the sleeve. Transfer the sleeve's first motion to the blocker ring, so the ring hits the cone before the sleeve reaches the gear's teeth.



SYNCHRONIZER ENGAGEMENT SEQUENCE — the sleeve engages the gear's dog teeth, never the cone itself

> **REFRAME** The synchronizer isn't "synchronizing the sleeve to the gear." The sleeve/hub are already at shaft speed — friction is bringing the **freewheeling gear** up or down to match. A question can exploit that distinction.

Material note: paper-coated steel blocker rings wear fast with the wrong fluid — another reason manual-trans fluid type is a real diagnostic item, not "gear oil is gear oil."

2.1 SHIFT MECHANISM

The shift mechanism transfers the driver's lever motion to the shift forks. **Fork-to-sleeve groove clearance matters** — if the fork or groove wears on the leading edge, the fork can't push the sleeve far

enough to fully seat the dog teeth, and a partially engaged gear (dog teeth are angled like arrowheads) pops back out under load.

Shift shafts are typically named by the gear pairs they control: **1–2**, **3–4**, **5–R**. Older designs may use external linkage instead of internal rails.

2.2 DETENTS & INTERLOCKS

- **Detent** — spring-loaded ball that holds a shift rail/fork in position. Can be overcome by force; it holds the gear, it doesn't lock it rigidly.
- **Interlock** — mechanically prevents two gears being selected at once (commonly pin-and-ball). A mechanical-integrity feature, not a shift-feel feature.
- **Reverse lockout** — specifically prevents accidental reverse engagement.

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03 SYMPTOM DIAGNOSIS & TRANSAXLE CONCEPTS

3.1 THREE CLASSIC GEAR/SHIFT SYMPTOMS

SYMPTOM	WHAT'S ACTUALLY HAPPENING
Grinding / clash	Sleeve grinds on dog teeth that can't engage cleanly. **One gear** → that gear's blocker ring. **Every gear** → suspect clutch drag before condemning every synchro at once. Damage shows on the dog teeth, not the tooth flanks.
"Block out" — won't shift at all	Shifts fine engine-off, not running → input shaft is still turning when it shouldn't be (clutch drag / speed mismatch). Ask: is the synchronizer assembly truly stopped, or still rotating?
Pops out of gear	Fork-to-sleeve groove clearance (top answer) → shaft endplay → worn dog teeth → weak detent → linkage wear.

> **RULE** "Block out" and all-gear grinding are frequently **clutch** problems wearing a transmission costume. Rule out clutch drag before opening the case.

3.2 NOISE DIAGNOSIS — THE ROAD-SPEED TEST

Hold a constant road speed and shift through the gears, listening to how the pitch changes:

- Pitch **tracks road speed** in every gear (same in 3rd, 4th, 5th) → the noisy part turns at **output-shaft speed** → suspect the output shaft/bearing.
- Pitch is **different per gear** at the same road speed → the part's speed is changing with ratio → suspect the **gearset** engaged in that range.

Master question, here and everywhere else in A3: **what's rotating, and at what speed, when the noise happens?**

3.3 TRANSAXLE-SPECIFIC CONCEPTS

In a transaxle, the **input shaft ("clutch shaft")** is integral to the mainshaft assembly — not a separate piece. That's why it needs **no pilot bearing**: the whole assembly is self-supported by its own bearings. Compare a longitudinal (non-transaxle) transmission, where the input shaft is separate from the mainshaft and needs a **third bearing** — the pilot bearing, supported at the crank/flywheel — for that free end.

Because A3 treats the drivetrain as one system, a transaxle's final-drive output gets the same diagnostic treatment as a differential's output — this carries straight into Unit 3.

3.4 TORQUE STEER

Pulling to one side under hard acceleration. **Most common cause: unequal CV joint operating angles** between the two half-shafts (contributing factors: unequal shaft length, a longer shaft twisting more under load).

> **MENTAL MODEL** A wobble socket on an extension still transmits torque at an angle, but loses efficiency as the angle grows — past some point it whips and stops working. Equal angles on both sides cancel out symmetrically, even with imperfect mounts. Conceptual cousin: **bump steer** in suspension geometry.

QUICK REFERENCE / ONE PAGE

UNIT 2 CRAM SHEET

RATIOS

Reduction = more torque, less speed (numerically high). Overdrive = less torque, more speed (<1). Direct = 1:1.

Idler changes direction only — never the ratio, even at a different size. **`10:7:20` → `2:1`** overall.

CONSTANT MESH

Never fixed-to-fixed — always fixed gear ↔ speed gear (freewheeling until locked). Only one gearset locked at a time.

Helical gears: quiet, but cause endthrust → need thrust surfaces. Splash lubricated: countershaft throws oil onto mainshaft.

SYNCHRONIZER CHAIN

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FORK → SLEEVE → KEYS → RING → CONE
→ DOG TEETH (lock)
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Hub = fixed to shaft. Ring = blocks until speed matches. It syncs the **freewheeling gear** to hub speed, not "sleeve to gear." Wrong fluid → fast ring wear.

SHIFT MECHANISM

Detent = holds gear selected, can be overcome by force. Interlock = physically prevents two gears engaging at once.

Fork-to-sleeve groove clearance must be right or the sleeve won't fully seat.

SYMPTOM LOGIC

One gear grinds → that synchro. All gears grind → clutch drag first.

Won't shift, but fine engine-off → clutch drag. Pops out → fork/sleeve clearance → endplay → dog teeth → detent → linkage.

NOISE TEST

Constant road speed, shift through gears:

Same pitch every gear → output shaft/bearing. Pitch changes per gear → that gearset.

Always ask: what's rotating, at what speed, when the noise happens?

TRANSAXLE

Input shaft = "clutch shaft," part of mainshaft → no pilot bearing needed.

Longitudinal trans: input shaft separate → needs pilot bearing (3rd support point).

Transaxle final-drive output = treat like a differential (bridges into Unit 3).

TORQUE STEER

Most common cause: unequal CV joint operating angles (also unequal shaft length/twist).

Equal angles both sides → no torque steer, even with imperfect mounts. Cousin: bump steer.